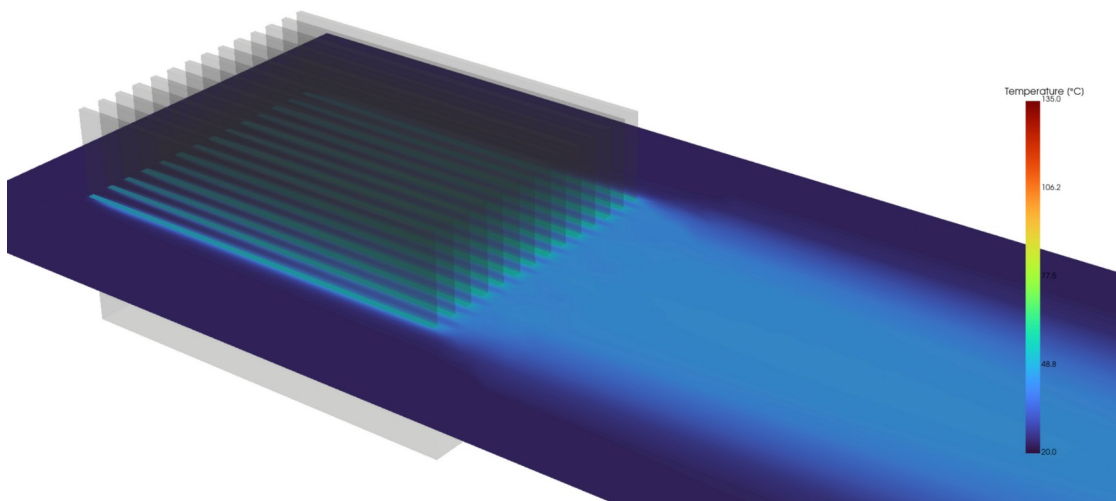




GRADIENT DYNAMICS

Conjugate Heat Transfer: Rectangular Fins - Gradient Dynamics Solver Validation Study



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1. Introduction

Effective thermal management is essential to the performance and service life of electronic components. Air-cooled plate-fin heat sinks remain widely used because they combine simple, manufacturable geometry with a large heat-transfer area and low operating cost. Their apparent simplicity, however, conceals a strongly coupled thermal-fluid problem: heat must spread from a small electronic package through the heat-sink base and fins before being removed by forced air flow. Fin efficiency, developing boundary layers, narrow inter-fin passages and flow bypass all influence the resulting junction temperature and overall thermal resistance [1].

This study validates the Gradient Dynamics FluxCore solver for conjugate heat-transfer (CHT) problems. The objective is to demonstrate that FluxCore can reproduce the coupled conduction and forced-convection behaviour of a representative electronics-cooling assembly, rather than only the fluid flow or solid temperature field in isolation. The target case is based on the experimental characterization and thermal model reported by Ventola et al. [1].

The benchmark consists of a power transistor mounted beneath an extruded-aluminium plate-fin heat sink in a forced-air duct. Seven operating points span the tested range of approach velocity, inlet temperature and dissipated power. For each condition, FluxCore will predict the coupled air velocity and temperature fields, conduction through the chip and heat sink, and the junction temperature. The primary validation quantity is the junction-to-ambient thermal resistance, $R_{ja} = (T_j - T_a)/P$, which will be compared with the published experimental values and the SimScale reference solution. This is a demanding CHT benchmark because the solution must preserve heat flux across multiple solid-fluid interfaces while resolving large differences in material conductivity, a compact heat source, one-millimetre fins, 2.1 mm fin spacing, wake development and bypass flow around an unshrouded heat sink. Small errors in interface coupling, near-wall resolution or flow distribution can produce a measurable change in T_j and R_{ja} . At the same time, the published geometry, material properties and seven experimental operating points make the case reproducible and provide a direct, quantitative test of FluxCore's multi-region CHT capability.

2. Simulation Setup

2.1 Geometry and computational domain

The geometry follows the SimScale reconstruction [2] of the commercial heat sink tested by Ventola et al. [1]. The extruded-aluminium heat sink is 57.2 mm long and 41.4 mm wide, with a base thickness of 8.4 mm. Fourteen rectangular fins are each 21.8 mm high and 1.0 mm thick, with 2.1 mm clear spacing between adjacent fins. The heat source represents an STMicroelectronics STP130NS04ZB power transistor. Its modelled envelope is 15.5 x 10 x 4.5 mm and the contact area with the heat sink is 1.555 cm².

The heat sink and chip are placed within an air enclosure. Measured from the heat sink, the inlet section extends 6L upstream and the outlet section extends 15L downstream, where $L = 57.2$ mm is the heat-sink length. A symmetry plane through the longitudinal centreline is used for the fluid, heat sink and chip, so only half of the physical assembly is simulated. This retains the relevant fin-channel, wake and bypass-flow physics while reducing computational cost [2].

2.2 Computational mesh

The study mesh was generated using Gradient Dynamics' proprietary Cartesian, structured meshing engine, following the standard workflow for electronics simulations. It contains 3,005,974 cells, of which 1,859,477 are in the fluid region and 1,146,497 in the solid domain.

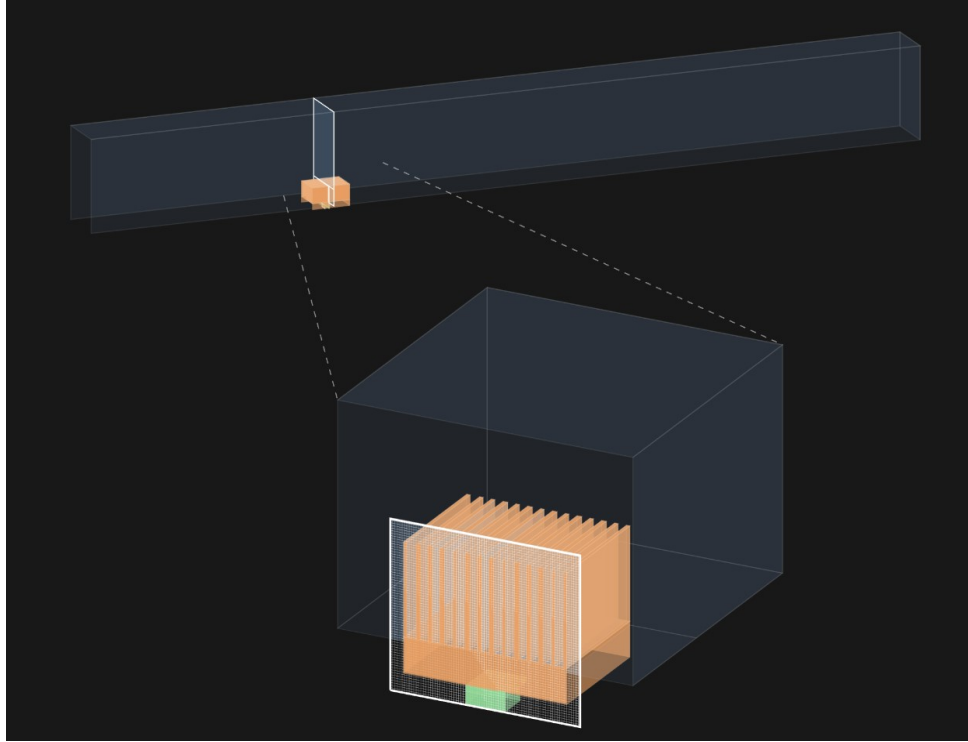


Figure 1. Mesh slice.

2.3 Materials and thermal coupling

Air is assigned a density of 1.179 kg/m^3 , kinematic viscosity of $1.529 \times 10^{-5} \text{ m}^2/\text{s}$, specific heat capacity of 1013 J/(kg K) , thermal-expansion coefficient of $3.43 \times 10^{-3} \text{ K}^{-1}$, laminar Prandtl number of 0.713 and turbulent Prandtl number of 0.85. The reference temperature is 273.1 K. The aluminium heat sink uses thermal conductivity $k = 209 \text{ W/(m K)}$, specific heat capacity $c_p = 897 \text{ J/(kg K)}$ and density 2700 kg/m^3 . The chip uses $k = 38.6 \text{ W/(m K)}$, $c_p = 705 \text{ J/(kg K)}$ and density 2330 kg/m^3 [2].

The chip conductivity is the effective value used in the SimScale model to represent the manufacturer's junction-to-case resistance of 0.5 K/W over the stated contact area [2]. At coupled chip-to-sink and sink-to-air interfaces, temperature and normal heat flux are continuous. This allows the calculation to capture heat spreading through the base, conduction along the fins and convective removal by the air in one energy-conserving CHT model.

2.4 Boundary conditions and operating points

Each case is run to a steady thermal and flow state. A uniform volumetric-flow inlet is prescribed with the corresponding inlet temperature; the downstream boundary is a 0 Pa gauge-pressure outlet. All solid surfaces exposed to the air are no-slip walls, the centre plane is a symmetry boundary and external enclosure walls are treated as adiabatic. The specified power is applied to the chip as the heat input. The seven conditions below reproduce the SimScale validation matrix [2].

Approach velocity [m/s]	Volumetric flow [m ³ /s]	Inlet temperature [K]	Chip power [W]
5.47	0.02700	296.9	56.64
7.03	0.03520	297.4	71.40
8.59	0.04295	297.9	82.36
9.96	0.04980	298.3	87.32
11.23	0.05620	298.9	85.07
12.50	0.06200	299.3	76.30
13.57	0.06783	299.6	60.24

Table 1. FluxCore operating points based on the SimScale rectangular-fin CHT validation case [2].

3. Results

FluxCore results are compared with the experimental measurements and the SimScale reference across all seven operating points. Figures 2 and 3 show the predicted junction temperature and junction-to-air thermal resistance as functions of approach velocity; Figures 4 and 5 show representative velocity and temperature fields.

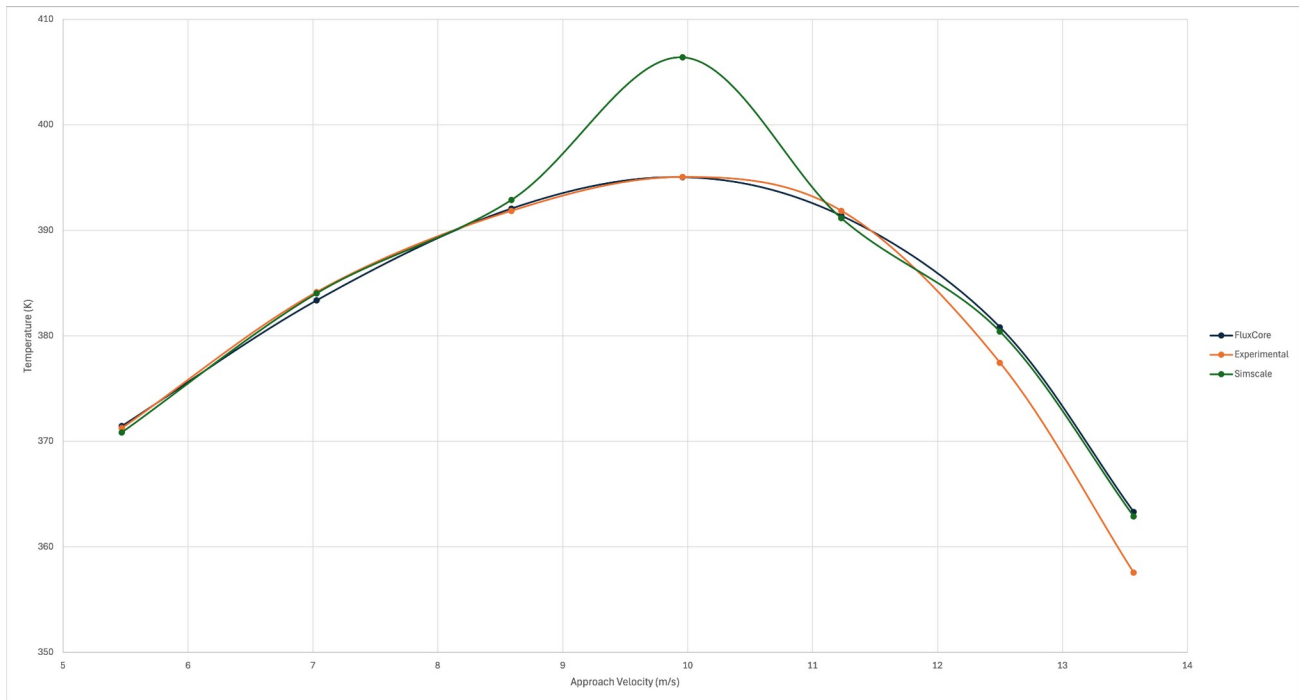


Figure 2. Junction temperature versus approach velocity.

FluxCore reproduces the non-monotonic junction-temperature response and tracks the experimental curve closely from 5.47 to 11.23 m/s, including the peak near 10 m/s. Unlike the SimScale result, which shows a pronounced overprediction at this operating point, FluxCore remains close to the measured value. At 12.50 and 13.57 m/s, FluxCore predicts higher junction temperatures than the experiment, although it remains close to SimScale.

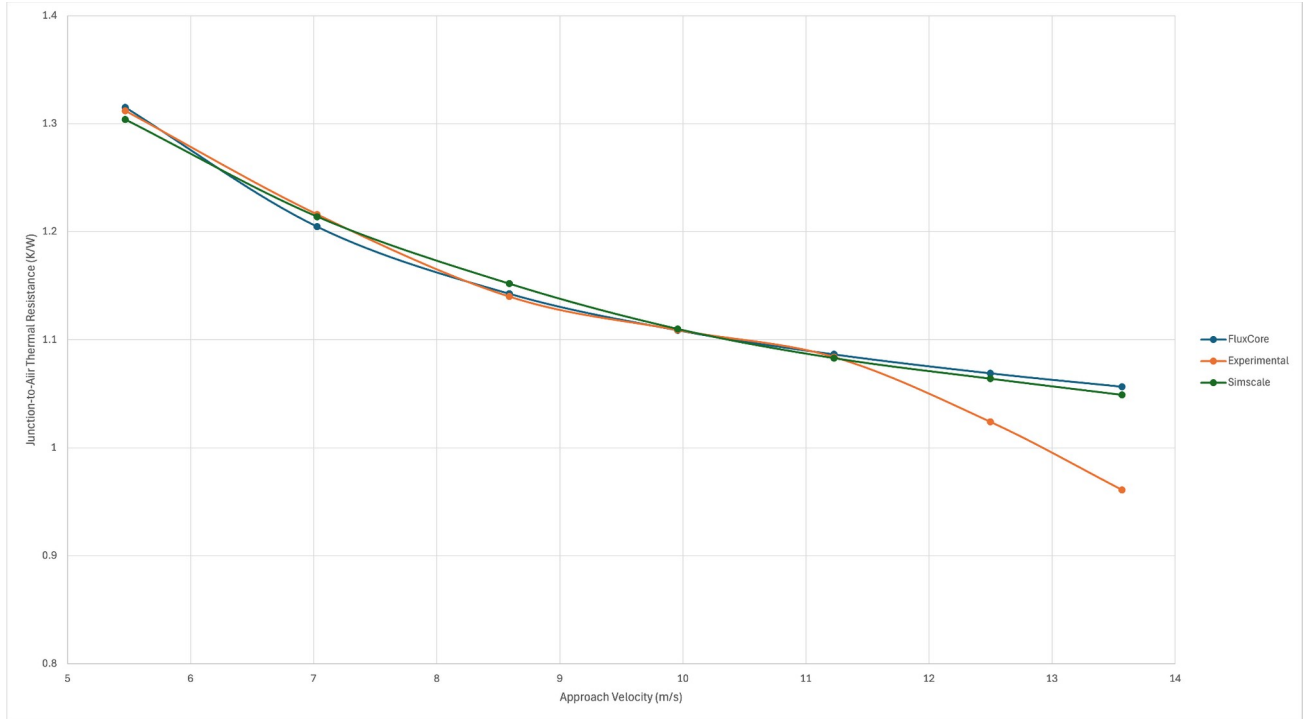


Figure 3. Junction-to-air thermal resistance versus approach velocity.

The thermal-resistance comparison leads to the same conclusion. FluxCore captures the expected reduction in R_{ja} as airflow increases and agrees closely with the experiment over the first five operating points. The largest departures occur at 12.50 and 13.57 m/s, where both CFD solutions remain near one another but exceed the experimental resistance. The common departure at these two points identifies the high-flow regime as the priority for sensitivity checks on boundary conditions, unmodelled heat-loss paths, material properties and mesh resolution.

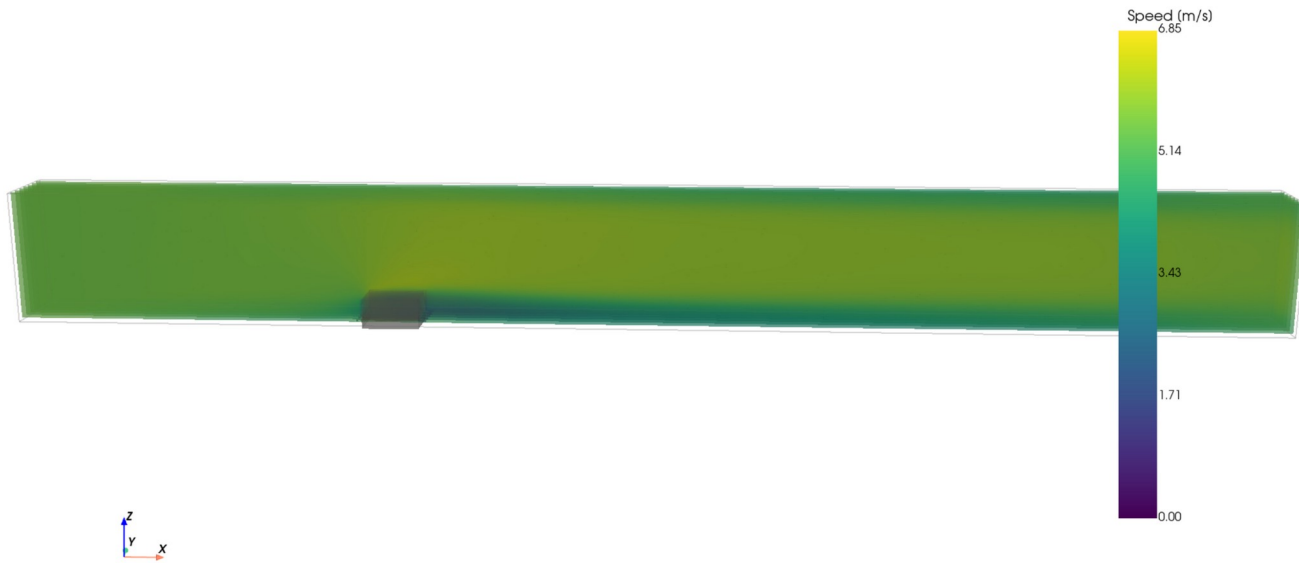


Figure 4. Velocity magnitude on a two-dimensional slice through the domain.

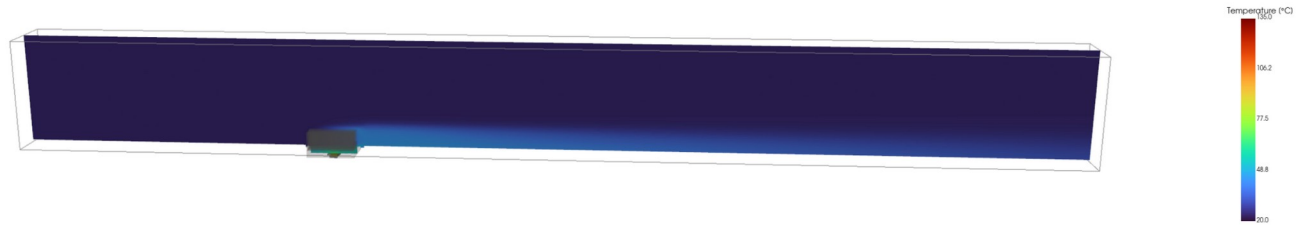


Figure 5. Temperature on a two-dimensional slice through the domain.

The field plots are consistent with the integrated results. The velocity slice shows the flow accelerating around the heat-sink region and a lower-speed wake downstream, while the temperature slice shows heat localized around the chip and base before being carried downstream in a decaying thermal plume. These features indicate that the solution is coupling solid conduction, forced convection and downstream heat transport in the expected manner.

4. Summary

FluxCore reproduces the published thermal response of the rectangular-fin heat sink across the seven operating points. It captures the measured junction-temperature peak near 10 m/s and the decrease in junction-to-air thermal resistance with increasing airflow, with close agreement over the low-to-intermediate velocity range.

At the two highest velocities, FluxCore and SimScale remain mutually consistent but both predict higher resistance and junction temperature than the experiment. Overall, the results support FluxCore's capability for steady multi-region CHT simulations involving localized heat generation, conduction through multiple solids and forced-air cooling, while clearly identifying the high-flow regime as the main area for further validation.

References

- [1] Ventola, L., Curcuruto, G., Fasano, M., Fotia, S., Pugliese, V., Chiavazzo, E. and Asinari, P. (2016). [Unshrouded Plate Fin Heat Sinks for Electronics Cooling: Validation of a Comprehensive Thermal Model and Cost Optimization in Semi-Active Configuration](#). *Energies*, 9(8), 608.
- [2] SimScale GmbH. [Validation Case: Conjugate Heat Transfer - Rectangular Fins](#). Last updated 8 February 2026.